



Started on	Tuesday, 15 September 2026, 9:24 AM
State	Finished
Completed on	Tuesday, 15 September 2026, 9:30 AM
Time taken	5 mins 29 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00**Problem Statement:**

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2  int findFloor(int arr[], int low, int high, int x) {
3      if (low > high)
4          return -1;
5      if (x >= arr[high])
6          return arr[high];
7      int mid = (low + high) / 2;
8      if (arr[mid] == x)
9          return arr[mid];
10     if (mid > 0 && arr[mid - 1] <= x && x < arr[mid])
11         return arr[mid - 1];
12     if (x < arr[mid])
13         return findFloor(arr, low, mid - 1, x);
14     return findFloor(arr, mid + 1, high, x);
15 }
16 int main() {
17     int n, x;
18     if (scanf("%d", &n) != 1) return 0;
19     int arr[n];
20     for (int i = 0; i < n; i++) {
21         scanf("%d", &arr[i]);
22     }
23     scanf("%d", &x);
24     printf("%d\n", findFloor(arr, 0, n - 1, x));
25     return 0;
26 }
```

	Input	Expected	Got	
✓	6 1 2 8 10 12 19 5	2	2	✓
✓	5 10 22 85 108 129 100	85	85	✓

	Input	Expected	Got	
✓	7	9	9	✓
	3			
	5			
	7			
	9			
	11			
	13			
	15			
	10			

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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